RANK-EIGHT GROTHENDIECK RINGS THROUGH MULTIPLICITY TWO: CRITERIA, MODELS, AND THE UNRESOLVED FRONTIER

Abstract. We analyze general complex categorification of every rank-eight fusion ring of multiplicity at most two in a complete census. An ordered collection of exact tensor and arithmetic tests excludes 832 of the 973 rings. Full multiplication-tensor comparisons identify unitary categories for 60 more. The remaining 81 rings are recorded as unresolved, including 3 already at multiplicity one; thus this is an explicit categorification frontier, not a complete classification. A rank-eight Wedderburn lemma extends the formal-codegree d-number obstruction to noncommutative rings. Every decision and every positive tensor comparison accompanies the manuscript in a checked machine-readable ledger.

1. Introduction and exact scope

The fusion-ring census [11] supplies all 96 rank-eight based fusion rings of exact multiplicity one and all 877 of exact multiplicity two. A fusion ring need not be the Grothendieck ring of a fusion category. We work over $\mathbb{C}$ and do not assume pivotality, sphericality, braiding, commutativity or unitarity when excluding a ring. A positive construction is a unitary category with the exact based-isomorphism type of its full multiplication tensor.

The lower-rank criteria and their finite verifiers are developed in [12, 13, 14, 15, 16, 17]. We cite their theorems and do not reprove them. These new manuscripts are cited by title, with no byline. Our rank-eight observation, Proposition 3.1, extracts ordinary formal codegrees from the Casimir spectrum even though the ring may be noncommutative. Known near-group, zero-spectrum and cyclotomic tests provide other general obstructions [24, 27, 20]. Standard quantum group, pointed, Tambara–Yamagami and representation categories supply the positive side [28, 40, 21, 41].

The accepted study of integral Grothendieck rings [18, 19] supplies a necessary divisibility test and explicit group-theoretical rank-eight models and exclusions. Being an integral or Drinfeld ring does not itself supply a fusion category, so only its explicit categorical constructions enter our positive list.

Theorem 1.1 (Certified rank-eight frontier)**.** *For the complete census through multiplicity two, the following trichotomy is exhaustive as an* analysis status*.*

Exact multiplicity	*Fusion rings*	*Excluded generally*	*Unitarily realized*	*Undecided generally*
1	96	45	48	3
2	877	787	12	78
Total	973	832	60	81

Every excluded ring is ruled out over $\mathbb{C}$ *with no pivotality assumption. Each positive model is matched to the entire tensor under a unit-preserving permutation, and is therefore a Grothendieck ring. For the* 81 *undecided rings, neither existence nor nonexistence is asserted. In particular, these numbers are* certified lower bounds *of* 48 *and* 12 *for the number of Grothendieck rings at the two multiplicities, rather than total counts of Grothendieck rings.*

The earliest undecided tensors are now `R8_1_9` and `R8_2_24`. The former first open ring `R8_1_3`, of global FP dimension 20, is realized below by a twisted group-theoretical category. Its dimension vector agrees with that of the commutative ring $\mathrm{Rep}(\mathrm{Dic}_5)$, although its multiplication is noncommutative. Thus full tensor comparison remains essential even after a suitable dimension profile is found.

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2. Trusted input and stable records

A rank-r fusion ring has a distinguished basis $B = (b_0 = 1, b_1, \dots, b_{r-1})$, structure constants $N_{ij}^k \in \mathbb{Z}_{\geq 0}$, and a duality involution $i \mapsto i^*$ satisfying

$$b_i b_j = \sum_k N_{ij}^k b_k, \qquad N_{ij}^0 = \delta_{j,i^*}, \qquad N_{ij}^k = N_{i^*k}^j = N_{kj^*}^i.$$

Its exact multiplicity is $\max_{i,j,k} N_{ij}^k$. A based isomorphism permutes the basis, fixes b_0, and preserves every structure constant; it automatically preserves duality. We do not identify tensors by character table, spectrum or FP dimension vector alone.

The source is the public *Fusion Ring Census* [11] at Git commit `b86eb6dc6eeeafe5fc5f6906d7b074e0322e9726`. It contains `results/rank8/tables.txt.gz` and `results/rank8/counts.csv`, with audit file `verification/rank8/audit.json`. The SHA-256 of the uncompressed tensor file is `97b86d2de8ca0352c5c6e14686cd1d43dbd52180dc53b2ff1ccf7a61788c9600`. The census verifies all duality types, associativity, nonnegative coefficients and uniqueness; its terminated-strata evidence justifies completeness through multiplicity two. We trust that source for counting and do not claim an independent exhaustive enumeration here.

Within each exact multiplicity m, sort full flattened $8 \times 8 \times 8$ tensors lexicographically and call the jth one $R8_m_j$. This is a deterministic local index, not an invariant of its original basis labelling. The files `data/decisions.json`, `data/realized_models.json` and `data/rank8_tables.txt.gz` give, respectively, one decision per input, the exact positive model tensors and the source multiplication tables. A SHA-256 of each JSON tensor prevents accidental source-record substitution. The comparison code minimizes the full tensor under all relabellings compatible with inexpensive intrinsic signatures; this is an isomorphism comparison, not a filter by numerical fingerprint.

3. Necessary criteria and their exact application

3.1. **Short tensor and subring tests.** If X is self-dual and $X^2 = 1 + kX$ with $k \geq 2$, its span with 1 is a fusion subring that cannot arise over $\mathbb{C}$ by [12, 22]. When exactly seven basis elements are invertible and one additional object X satisfies

$$gX = Xg = X, \qquad X^2 = \sum_{g \in G} g + \kappa X,$$

the rank-eight near-group case with $\kappa = 1$ or 2 is commutative and satisfies $0 < \kappa < |G| - 1$. Schopieray's general Proposition 3.2 [24] excludes it. The machine certificate checks the complete group and all displayed equations, not just the value of κ.

The zero-spectrum test is the finite nine-index condition of [27, Theorem 2.14] in the form reproduced in [16, Section 3]. It is a necessary condition for an associative tensor lift, with no pivotality assumption. Its C++ scanner searches the nine indices on the full tensor and the Python checker independently recomputes every coefficient, positivity condition and vanishing contraction. The native arrays are sized for rank eight, and the checker has no rank-six restriction.

Every duality-closed, multiplication-closed based subset of a category gives a fusion subcategory. We compare every such proper subring of ranks two through six against the *complete positive list* in the corresponding lower-rank theorem. The valid bounds are unrestricted for rank two and three, 128 for rank four, 20 for rank five, and 2 for rank six [12, 13, 14, 15, 16]; every rank-eight subring in this manuscript also has multiplicity at most two. A rank-seven subring is not excluded merely because it is absent from an incomplete rank-seven list.

The rank-seven analysis [17] provides four specific, fully checked negative multiplicity-two tensors that occur as rank-eight closed subrings. We transfer their exact signed-sphericalization witnesses by complete seven-dimensional based isomorphism and recheck all four proof trees. This separate

stage gives exclusions only for rings actually containing one of those four proved negative subrings; it makes no claim that the rank-seven positive catalogue is exhaustive.

The involution-module theorem of [16, Section 5] says that an invertible involution fixing basis elements forces a finite anti-invariant module representation. Where its stated hypothesis is met, the script searches *all* bounded integer matrices and verifies multiplication, transpose, parity and entry bounds. The FP dimension quotient of a hypothetical fusion subcategory must likewise be an algebraic integer [21, Proposition 8.15]; a rational coefficient in its exact minimal polynomial excludes categorification.

3.2. **A rank-eight noncommutative codegree lemma.** For any fusion ring, write

$$\Omega = \sum_i b_i b_{i^*}, \qquad C_R = \sum_i N_i N_i^{\mathsf{T}}.$$

For an irreducible representation E of $R \otimes \mathbb{C}$ of degree e_E, the central element Ω acts as the scalar $e_E f_E$, where f_E is its ordinary formal codegree. The regular representation contains E with multiplicity e_E, so that scalar has multiplicity e_E^2 in the characteristic polynomial of C_R [23, 16].

Proposition 3.1 (Noncommutative rank-eight codegrees)**.** *Let R be a noncommutative fusion ring of rank eight. Its semisimple complex algebra has Wedderburn decomposition $\mathbb{C}^4 \oplus M_2(\mathbb{C})$. The formal codegree of the unique matrix-block representation is a rational integer. Every irreducible factor of the characteristic polynomial of C_R of degree at least two consists entirely of formal codegrees of one-dimensional representations. Consequently, if any such factor fails Ostrik's d-number divisibility condition, R has no complex fusion categorification.*

Proof. The FP dimension character is one-dimensional, so at least one scalar summand exists. Noncommutativity requires a matrix summand of dimension at least two. Since the sum of the squares of the degrees of simple modules is eight, there can be only one matrix summand and it must have degree two; the four remaining dimensions are one. The rational based algebra is split by some number field. Galois automorphisms permute its irreducible representations preserving their degrees, so the unique two-dimensional representation is Galois-fixed. Its formal codegree is an algebraic integer by [23] and rational by Galois invariance, hence belongs to $\mathbb{Z}$. Its Casimir eigenvalue is therefore rational. Every nonlinear irreducible factor of the integral Casimir polynomial comes from the four one-dimensional representations; on each, the eigenvalue is exactly the formal codegree. Finally, formal codegrees of complex fusion categories are d-numbers [23]. For an irreducible polynomial $q(t) = t^d + a_1 t^{d-1} + \cdots + a_d$, this requires $|a_d|^j \mid a_j^d$ for every $j = 1, \ldots, d$. Failure of one exact integer divisibility gives the assertion. □

The commutative branch uses the ordinary d-number test on every Casimir factor. The script never treats raw Casimir eigenvalues as formal codegrees of noncommutative representations without Proposition 3.1. We also ran the rational-indivisibility screen of [25, Theorem 4.4], noncommuting field-automorphism witnesses from [16], and finite-field cyclotomic witnesses from [14]. These three screens add no exclusions in the present bound; their zero rows and exact code remain in the stage data. Schopieray's greedy formal-codegree obstruction $2, 3, 7$ [26] cannot occur before multiplicity five, since its forced rank-four subring has a coefficient five, so it is not an active step here.

3.3. **Integral divisibility and the complete signed quotient test.** The integral Drinfeld condition [18, Section 2.3] says that, if all simple FP dimensions of a categorifiable ring are integers, every ordinary formal codegree is a positive integer dividing its global FP dimension. For a commutative tensor, the verifier finds its unique positive FP character by an exact integral eigenvector of the strictly positive matrix $\sum_i N_i$. It then compares every exact linear factor of $\det(tI - C_R)$ with $D = \sum_i \mathrm{FPdim}(b_i)^2$. A nonintegral factor, or an integer codegree that does not divide D, is an unconditional complex no-go. This stage includes concrete multiplicity-two failures with $D = 156$ and 110, respectively, at $R8_2_288$ and $R8_2_392$.

The archived rank-eight appendix of [19] prints full tensors and explicit categorification decisions. Its items C.3.2(5) and C.3.2(7) are excluded by the paper's no-induction-matrix Lemmas 4.3 and 4.2. The appendix-to-census verifier checks all 8^3 entries under a unit-preserving permutation before transferring either published exclusion. We rely on those cited lemmas for the induction obstruction; their original Normaliz computation is not repeated here. The same transfer finds group-theoretical models in items C.3.2(3), C.3.2(6), and C.3.2(11). Item C.3.2(4) matches $R8_1_3$ but supplies no existence or exclusion claim; the explicit twisted group-theoretical construction below now realizes that ring.

Finally, the oriented signed-sphericalization theorem of [16, 17] applies even when the alleged category has no pivotal structure. The canonical sphericalization has two lifts of each simple; quotienting its Grothendieck ring by the negative central invertible yields an ordered signed multiplication tensor. We enumerate every cyclic and duality orbit of signed coefficients within the exact parity bounds, require all ordered associativity equations, then check every possible spherical character by the inherited norm, duality, and dimension-order criteria. A finite search that times out or has an unrefuted branch supplies no exclusion. The attached campaign records each attempted multiplicity-one tensor, incomplete result, and timeout separately; only certificates whose complete signed branches independently regenerate are admitted to the ledger.

3.4. **Grading obstructions with rank-one components.** The following short arguments give five further exclusions without any pivotality assumption. In a faithful grading $\mathcal{C} = \bigoplus_{g\in G} \mathcal{C}_g$, every component is an invertible bimodule category over $\mathcal{D} = \mathcal{C}_e$, and $g \mapsto [\mathcal{C}_g]$ is a homomorphism to $\mathrm{BrPic}(\mathcal{D})$ [7, Theorem 1.3 and Section 6].

Lemma 3.2. *If a pointed category* Vec_A^ω *has a module category with one simple object, then* $[\omega] = 0$.

Proof. Identify the module category with Vec. Each invertible object acts by the identity functor, up to isomorphism. The scalar module constraints form a nonzero 2-cochain whose coboundary is ω, by the module pentagon. Hence the associator class is trivial. □

Proposition 3.3. *The rings* $R8_1_26$*,* $R8_2_116$*, and* $R8_2_189$ *have no complex fusion categorification.*

Proof. In the original zero-based census labels, $R8_1_26$ has a faithful C_4 grading with components

$$\{0,3,4\}, \quad \{1\}, \quad \{5,6,7\}, \quad \{2\}.$$

Its neutral group is $A = C_3$. Lemma 3.2 identifies its neutral category with Vec_{C_3}. By [7, Corollary 1.2], its Brauer–Picard group is the orthogonal group of $A \oplus A^*$ with hyperbolic quadratic form. Over $\mathbb{F}_3$ this consists of $I, -I, J, -J$, where J exchanges the two coordinates. Every square is the identity, so $[\mathcal{C}_2] = [\mathcal{C}_1]^2 = 1$. On the other hand, the degree-two invertible object $s = 5$ conjugates $a = 3$ to $a^{-1} = 4$. Its component is the regular left module with right action twisted by inversion. It cannot be equivalent to the regular bimodule: such an equivalence would send s to some $b \in A$ and the relation $as = sa^{-1}$ would imply $ab = ba^{-1}$, which is false in C_3.

For $R8_2_116$ and $R8_2_189$, the universal grading has five components, of ranks $4, 1, 1, 1, 1$, so its group is C_5. Their neutral groups are respectively C_4 on $\{0, 1, 2, 7\}$ and C_2^2 on $\{0, 5, 6, 7\}$. Again their associator classes are trivial. In the first case the Brauer–Picard group embeds in $\mathrm{GL}_2(\mathbb{Z}/4)$, of order 96; in the second it is $O_4^+(\mathbb{F}_2)$, of order 72. Neither group has an element of order five. Hence the component homomorphism is trivial, forcing every component to be equivalent to the rank-four regular bimodule, contrary to the rank-one components. The verifier enumerates the small orthogonal groups and checks every grading support identity exactly. □

Proposition 3.4. *The rings* $R8_2_78$ *and* $R8_2_168$ *have no complex fusion categorification.*

Proof. Each has a faithful grading with a rank-one component and neutral fusion rules $\mathrm{Rep}(S_3)$: $g^2 = 1$, $gx = xg = x$, and $x^2 = 1 + g + x$. A rank-one module category gives a fiber functor

on the neutral category $\mathcal{D}$, so $\mathcal{D} \simeq \mathrm{Rep}(H)$ for a semisimple Hopf algebra of dimension six. The dimension-six classification [10] and the rank-three fusion rules force $H \simeq \mathbb{C}[S_3]$. The dual category of its fiber-functor module is $\mathrm{Rep}(H^*) \simeq \mathrm{Vec}_{S_3}$, of rank six [21, Example 7.12.26]. But invertibility of the homogeneous $\mathcal{D}$-bimodule would identify that dual with $\mathcal{D}$, of rank three, a contradiction. □

3.5. **Solvability, equivariantization and module ranks.** We give three further general exclusions using only small groups and exact fusion rules. By [3, Theorem 1.6], a fusion category of FP dimension $p^a q^b$ is solvable. The characterization in its Definition 1.2 and Proposition 4.4 expresses it as successive cyclic-prime extensions and equivariantizations. If its adjoint subcategory is the whole category, its last nontrivial step cannot be an extension. Thus it is an actual C_p-equivariantization, with a central $\mathrm{Rep}(C_p)$ whose underlying invertibles embed into the category. We use the orbit/stabilizer description of equivariant simples [6, Section 2]; cyclic stabilizers have only one-dimensional irreducible projective representations.

Proposition 3.5. *The ring R8_1_29 has no complex fusion categorification.*

Proof. Its invertibles form C_6, its other two simples X, Y are self-dual of dimension three, and its adjoint ring is the whole ring. Its dimension is 24, so a putative category $\mathcal{C}$ is $\mathcal{D}^{C_p}$ with $p = 2$ or 3.

For $p = 3$, the order-three character subgroup fixes X, Y and has two free orbits on the six invertibles. The forgetful images of X, Y split into three distinct invertibles each; those of the invertibles give two more. Thus $\mathcal{D} = \mathrm{Vec}_A^\omega$, $|A| = 8$, with two free C_3-orbits. The only groups of order eight admitting an order-three automorphism are C_2^3 and Q_8. On C_2^3 the action has a fixed line and an irreducible plane. The three nonzero elements in that plane form a free orbit whose product has coefficient two into the same free orbit. The output stabilizer is trivial, so associator and tensorator choices cannot change this coefficient. On Q_8, inversion exchanges the orbits $\{i, j, k\}$ and $\{-i, -j, -k\}$; their two induced dimension-three objects are distinct duals. These conclusions contradict multiplicity one and self-duality, respectively.

For $p = 2$, the character involution exchanges X, Y. De-equivariantization has the fusion rules

$$G(\mathcal{D}) = C_3 = \langle a \rangle, \qquad a\rho = \rho a = \rho, \qquad \rho^2 = 1 + a + a^2 + 2\rho.$$

It has dimension twelve and trivial universal grading. The same last-step argument gives $\mathcal{D} = (\mathrm{Vec}_V^\omega)^{C_3}$, where $V = C_2^2$ and C_3 permutes the three nonzero elements transitively. No assumption on ω is made. A monoidal automorphism of $\mathrm{Id}_\mathcal{D}$ has scalar one on ρ, because ρ occurs in ρ^2, and then on all invertibles. Consequently the map from central invertibles to underlying invertibles is injective, and the central $\mathrm{Rep}(C_3)$ is unique.

The C_2 action on $\mathcal{D}$ fixes every simple: exchanging a, a^2 would produce a dimension-two simple upstairs. Every such tensor autoequivalence preserves the unique central $\mathrm{Rep}(C_3)$. Its restriction is tensor-isomorphic to the identity, since $H^2(C_3, \mathbb{C}^\times) = 0$. Equivariantization/de-equivariantization [4, Theorem 4.18(i) and Proposition 4.19] therefore lifts it to a C_3-equivariant tensor autoequivalence of Vec_V^ω. Its object permutation centralizes C_3 in $\mathrm{Aut}(V) = S_3$, so lies in C_3. Composing with the inverse action functor removes that permutation without changing the descended equivalence class.

Identity-on-objects tensor autoequivalences of Vec_V^ω are classified by $H^2(V, \mathbb{C}^\times) = C_2$: the two associators cancel in the tensorator equation. Their equivariant structures have ambiguity $H^1(C_3, \widehat{V}) = 0$, since multiplication by three is invertible on $\widehat{V}$. Both possible classes have explicit representatives. Besides the identity, take

$$J(u, v) = (-1)^{u_1 v_1 + u_1 v_2 + u_2 v_2}, \qquad u, v \in \mathbb{F}_2^2.$$

This nontrivial normalized 2-cocycle is C_3-invariant and satisfies $J^2 = 1$. Thus it commutes even with nontrivial scalar tensorators of the existing C_3 action and descends with identity comparisons. There are no further C_2 coherence choices, because $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{D}) = 1$.

The identity action leaves the coefficient two in ρ^2. For the J action, all degree-zero channels have $J(u, u) = -1$ for $u \neq 0$; for each nonzero output degree, the two ordered channels have opposite

signs. Hence each lift $\rho_\pm$ has square containing the three negative-character invertibles and one copy of each $\rho_\pm$. The unit occurs in $\rho_+\rho_-$, not in either square. The two lifts are distinct duals, again contradicting the self-duality in R8_1_29. This exhausts both prime branches for arbitrary pointed associators. □

Lemma 3.6 (Module-rank divisibility)**.** *For a fusion category $\mathcal{B}$, let $g_\mathcal{B}$ be the greatest common divisor of the ranks of its indecomposable finite semisimple module categories. If $\mathcal{B} \subseteq \mathcal{D}$ is a fusion subcategory, then $g_\mathcal{B} \mid \mathrm{rank}(\mathcal{D})$.*

Proof. Decompose $\mathcal{D}$, with its left tensor action by $\mathcal{B}$, into indecomposable module categories. Their ranks add to $\mathrm{rank}(\mathcal{D})$. This statement uses ranks in the specified module presentation; it makes no assertion that module ranks are Morita invariant. □

Proposition 3.7. *The rings R8_1_45 and R8_1_47 have no complex fusion categorification.*

Proof. Each ring has simple dimensions $(1,1,2,3,3,4,4,4)$ and admits labels $1, h, E, P, Q, W_0, W_1, W_2$ with

$$h^2 = 1, \quad hE = E, \quad hP = Q, \quad hW_i = W_i, \quad P^* = Q,$$
$$E^2 = 1 + h + E, \qquad EP = EQ = P + Q, \qquad PQ = 1 + E + P + Q.$$

The adjoint ring is the whole ring. Dimension 72 and the invertible group C_2 therefore force a putative categorification to be $\mathcal{C} = \mathcal{D}^{C_2}$ with $\mathrm{FPdim}(\mathcal{D}) = 36$. The forgetful images are

$$F(1) = F(h) = 1, \quad F(E) = a + a^2, \quad F(P) = F(Q) = Y, \quad F(W_i) = U_i + V_i.$$

Here a has order three, Y is simple of dimension three, and the six U_i, V_i are distinct simples of dimension two. Thus $\mathrm{rank}(\mathcal{D}) = 10$. The full fusion subcategory $\mathcal{B}$ on $1, a, a^2, Y$ satisfies

$$aY = Ya = Y, \qquad Y^* = Y, \qquad Y^2 = 1 + a + a^2 + 2Y.$$

The C_2 action exchanges a, a^2 and fixes Y; its equivariantization inside $\mathcal{C}$ has the displayed nonselfdual P, Q.

As in the preceding proof, solvability and trivial universal grading imply $\mathcal{B} = (\mathrm{Vec}_V^\omega)^{C_3}$, $V = C_2^2$, with the transitive action. The restriction map

$$H^3(V, \mathbb{C}^\times) \longrightarrow \prod_{\substack{L \le V \\ |L|=2}} H^3(L, \mathbb{C}^\times)$$

is an isomorphism $C_2^3 \to C_2^3$. In integral-cohomology generators u^2, v^2, uv, the three restrictions of (r,s,t) are $(r, s, r+s+t)$. Since C_3 permutes the factors, there are exactly two invariant classes: zero and the class nontrivial on every subgroup of order two.

Suppose first $[\omega] = 0$. The lift of the object C_3 action is unique in $\mathrm{Aut}_\otimes(\mathrm{Vec}_V) = C_2 \times S_3$, and its coherence is unique because $H^2(C_3, \widehat{V}) = 0$. Therefore $\mathcal{B} \simeq \mathrm{Rep}(A_4)$. Its central $\mathrm{Rep}(C_3)$ is unique, since $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{B}) = 1$, and is preserved by the C_2 action. Sequential equivariantization and de-equivariantization [3, Section 2.6 and Proposition 2.10][4, Proposition 4.19] identify $\mathcal{B}^{C_2}$ with $(\mathrm{Vec}_V)^{S_3}$. The action on nonzero elements is the usual faithful S_3 permutation action; inversion on the character C_3 forces the group $C_3 \rtimes C_2 = S_3$.

There are exactly two lifts of this object action to $C_2 \times S_3$, given by the trivial or parity homomorphism into $H^2(V, \mathbb{C}^\times) = C_2$. Their coherence ambiguity vanishes: $H^j(S_3, \widehat{V}) = 0$ for $j > 0$, by the normal C_3 subgroup, whose positive cohomology and fixed space on $\widehat{V}$ both vanish. Both lifts yield self-dual dimension-three orbit objects. For completeness choose a stabilizing transposition τ swapping coordinates and its fixed nonzero element $v = (1,1)$. For the trivial lift, the equivariance scalars $u = \pm 1$ act trivially on the unit channel of $v \otimes v$. For the other lift choose tensorator $\mu(a,b) = (-1)^{a_1 b_2}$. Its square is trivialized by $\eta_a = (-1)^{a_1 a_2}$, so $\eta_v = \mu(v,v) = -1$. The equivariance scalar satisfies $u^2 = \eta_v$ and acts on the unit channel by $u^2/\mu(v,v) = 1$. A τ-invariant character

change of η is one on v, so cannot change the conclusion. Frobenius reciprocity for the orbit induction gives a tensor unit in each of the two squares. This contradicts $P^* = Q$.

It remains that $[\omega]$ is nontrivial on every order-two subgroup. The crossed product of Vec_V^ω by C_3 is pointed, and realizes $\mathcal{B}$ as the group-theoretical category

$$\mathcal{B} \simeq \mathcal{C}(A_4, \Omega, K, \psi), \qquad K = C_3, \qquad [\Omega]|_V = [\omega], \quad [\Omega]|_K = 0.$$

By the classification of pointed module categories and its double-coset rank formula [5, Example 2.1, Theorem 3.1 and Proposition 3.1], indecomposable $\mathcal{B}$-module categories correspond to subgroups $H \leq A_4$ with $[\Omega]|_H = 0$ and permissible cochains. If $|H|$ were even, Cauchy's theorem would give a subgroup of order two, contradicting the restriction of Ω. Thus H is trivial or has order three. For $H = 1$, the four double cosets in $K\backslash A_4/H$ have trivial intersections, giving module rank four. For $H = C_3$, conjugate it to K. There are two double cosets, with intersections C_3 and 1; projective cocycles on C_3 are coboundaries, so their contributions are three and one. Every indecomposable $\mathcal{B}$-module category therefore has rank four, regardless of the permissible cochains. Lemma 3.6 would force $4 \mid \mathrm{rank}(\mathcal{D}) = 10$, a contradiction. □

The portable verifiers check the source-bound tensors, the displayed fusion identities, the complete small cocycle identities, and all relevant A_4 subgroups and double cosets. These finite checks certify the hypotheses; the categorical exhaustion is supplied by the proofs above. Neither argument assumes a braiding or a pivotal structure.

3.6. **Ordered exclusion ledger and the first open cases.** An excluded tensor is removed after the first verified certificate in the following order. For each row, "left" still counts rings subsequently recognized as positive, so this is an exclusion table rather than a table of outstanding questions.

Criterion	New $m = 1$	New $m = 2$	Left $m = 1$	Left $m = 2$
Near-group stabilizer	1	1	95	876
Rank-two closed subring	0	10	95	866
Zero-spectrum tensor condition	7	80	88	786
Proved rank 2–6 subring	3	121	85	665
Four proved rank-seven subrings	0	12	85	653
Commutative codegree d-number	23	462	62	191
Rank-eight noncommutative codegree	1	17	61	174
Integral formal-codegree divisibility	1	6	60	168
Published no-induction-matrix lemma	2	0	58	168
Involution-module obstruction	0	2	58	166
Field-automorphism obstruction	0	0	58	166
Fusion-subcategory dimension quotient	0	1	58	165
Primitive formal-codegree test	0	0	58	165
Cyclotomic eigenvalue test	0	0	58	165
Complete oriented signed quotient	3	71	55	94
Graded Brauer–Picard obstruction	1	2	54	92
Rank-one fiber-functor obstruction	0	2	54	90
Solvability and equivariantization	3	0	51	90

The first unresolved IDs are $R8_1_9$ and $R8_2_24$. Every negative stage is a theorem applied to tensor data or an exact based-tensor transfer of a published and proved no-go. Proposition 3.1 and the integral-divisibility stage both have certified first applications already at multiplicity one; the latter also resolves two multiplicity-two cases remaining open after the prior stages. These *first appearances* refer only to the two multiplicities of the complete rank-eight census. Nothing in this calculation establishes a multiplicity from which no new criterion is needed.

4. Exact positive models

Each standard family in this section admits a unitary complex realization. A pointed group ring of order eight is $K_0(\mathrm{Vec}_G)$ for the group reconstructed from its tensor. The five groups are C_8, $C_4 \times C_2$, C_2^3, D_8 and Q_8. The near-group ring $C_7 + 0$ is realized by a Tambara–Yamagami category [40]. If A is a rank-four unitary model from [14, Table 1] and B is either Vec_{C_2} or Fibonacci, the Deligne product $A \boxtimes B$ is a unitary rank-eight model with multiplication tensor $N_A \otimes N_B$. Its multiplicity is at most two precisely for the displayed matches.

The three archived models $\mathcal{C}(F_5, 1, D_5, 1)$, $\mathcal{C}(S_4, 1, A_4, 1)$, and $\mathcal{C}(\mathrm{PSL}(2,7), 1, C_7 \rtimes C_3, 1)$ are untwisted group-theoretical fusion categories, hence admit unitary realizations. Their exact appendix tensors are matched against the census in all 512 positions, with a stored basis permutation for every match [19, Appendix C.3.2(3), (6), (11)]. Some admit a second subgroup construction, also recorded in the model catalogue.

Proposition 4.1 (Representation and quantum-group templates)**.** *All three character-ring templates* $\mathrm{Rep}(D_{20})$*,* $\mathrm{Rep}(D_{26})$*, and* $\mathrm{Rep}(\mathrm{Dic}_5)$ *have rank eight and multiplicity one. The standard unitary* $SU(2)_7$ *category, and the even subcategories of* $SU(2)_{14}$ *and* $SU(2)_{15}$*, also have rank eight and multiplicity one. Their generated multiplication tensors occur in the supplied census and are matched by full based-ring isomorphisms.*

Proof. For $SU(2)_k$, the simples are the labels $0, \dots, k$ and

$$N_{ab}^c = 1 \quad \Longleftrightarrow \quad |a-b| \le c \le \min(a+b, 2k-a-b), \qquad a+b-c \equiv 0 \pmod 2;$$

all other coefficients are zero. The even labels close under multiplication and yield eight objects for $k = 14, 15$, while $k = 7$ already has eight labels [28]. The explicit tensor generator implements this formula directly.

For the dihedral group of order $2n$, take $n = 10$ or 13. Its two-dimensional characters ρ_j obey $\rho_i\rho_j = \rho_{i+j} + \rho_{|i-j|}$, with $\rho_0 = 1 + \beta$ and $\rho_{n-j} = \rho_j$. For $n = 10$, $\rho_5 = \alpha + \alpha\beta$, the four linear characters form C_2^2, $\beta\rho_j = \rho_j$ and $\alpha\rho_j = \rho_{5-j}$. For $n = 13$ there are two linear characters, with $\beta\rho_j = \rho_j$. Thus their ranks are respectively $4+4$ and $2+6$. For Dic_5 of order twenty the four linear characters χ_i form C_4, with

$$\rho_0 = \chi_0 + \chi_2, \quad \rho_5 = \chi_1 + \chi_3, \quad \rho_{10-j} = \rho_j, \quad \chi_i\rho_j = \begin{cases} \rho_j & i \text{ even}, \\ \rho_{5-j} & i \text{ odd}; \end{cases}$$

the same product $\rho_i\rho_j = \rho_{i+j} + \rho_{|i-j|}$ completes the ring. These are ordinary finite-group character identities [41]. The supplied generator expands each product and compares all 8^3 ordered coefficients after unit-fixing relabelling. A match is a ring isomorphism and the corresponding representation category supplies the construction. □

Multiplicity	Ring ID	One exact unitary model
1	`R8_1_1`	`Rep(Dic5)`
1	`R8_1_3`	$\mathcal{C}(C_5 \rtimes C_4, \omega_2, C_2, 1)$
1	`R8_1_4`	`Vec(G), |G|=8`
1	`R8_1_5`	`Vec(C4) x Fib`
1	`R8_1_6`	Bosonic quotient of $PSU(2)_6 \boxtimes \mathcal{C}(C_4, q)$
1	`R8_1_7`	Central square-root extension of pseudo-$PSU(2)_6$
1	`R8_1_10`	`Vec(G), |G|=8`
1	`R8_1_11`	Central-twisted flip extension of pseudo-$PSU(2)_6$
1	`R8_1_12`	Isotropic orbifold of $NG(C_{16}, 16)$
1	`R8_1_13`	`Vec(G), |G|=8`
1	`R8_1_14`	Central-twisted flip extension of $PSU(2)_6$
1	`R8_1_15`	$\mathrm{HI}(C_4)$

Multiplicity	Ring ID	One exact unitary model
1	R8_1_16	Vec(G), |G|=8
1	R8_1_19	$(\mathrm{Fib}\boxtimes\mathrm{Fib})\rtimes C_2$
1	R8_1_20	TY(C7)
1	R8_1_22	Diagonal parity in $\mathrm{TY}(C_3)\boxtimes\mathrm{Vec}_{C_4}$
1	R8_1_23	$\mathrm{Vec}_{Q_8}^{C_3}$
1	R8_1_24	C(F5,1,D5,1)
1	R8_1_25	$PSU(2)_6\rtimes C_2$ with the flip action
1	R8_1_27	TY(C3) x Vec(C2)
1	R8_1_30	$\mathrm{TY}(C_6)^{C_2}$ with odd coherence -1
1	R8_1_32	pseudo-PSU(2)_6 x Vec(C2)
1	R8_1_34	$\mathrm{TY}(C_9)^{C_2}$ with odd coherence -1
1	R8_1_35	$\mathrm{TY}(C_3^2)^{C_2}$ with odd coherence -1
1	R8_1_36	TY(C3) crossed C2
1	R8_1_39	TY(C3) x Fib
1	R8_1_42	Pseudo-$PSU(2)_6\rtimes C_2$ with the flip action
1	R8_1_43	pseudo-PSU(2)_6 x Fib
1	R8_1_53	Isotropic orbifold of $NG(C_4^2,16)$
1	R8_1_59	Rep(D10) x Vec(C2)
1	R8_1_61	One-half SO(12)_2
1	R8_1_65	Vec(G), |G|=8
1	R8_1_66	Vec(C2xC2) x Fib
1	R8_1_67	PSU(2)_6 x Vec(C2)
1	R8_1_69	$\mathrm{HI}(C_2^2)$
1	R8_1_70	Fib x Vec(C2) x Fib
1	R8_1_71	PSU(2)_6 x Fib
1	R8_1_72	PSU(2)_7 x Vec(C2)
1	R8_1_73	PSU(2)_14
1	R8_1_78	SO(9)_2
1	R8_1_79	Z(Vec(S3))
1	R8_1_83	Integral part of the $m=0$ triality gauging
1	R8_1_85	Neutral part of S3-gauged SO(8)_1
1	R8_1_91	Rep(D_{26})
1	R8_1_92	PSU(2)_15
1	R8_1_93	Rep(D10) x Fib
1	R8_1_94	PSU(2)_7 x Fib
1	R8_1_95	Fib x Fib x Fib
2	R8_2_39	Asaeda–Haagerup AH_4
2	R8_2_154	$\mathrm{TY}(C_2^2)^{S_3}$ with parity coherence
2	R8_2_174	Rep(A4) x Vec(C2)
2	R8_2_202	C(S4,1,A4,1)
2	R8_2_212	Rep(A4) x Fib
2	R8_2_241	$\mathcal{C}(\mathrm{PSL}(2,7),1,C_7\rtimes C_3,1)$
2	R8_2_477	even 2D2 x Vec(C2)
2	R8_2_484	Asaeda–Haagerup AH_5
2	R8_2_491	even 2D2 x Fib
2	R8_2_493	Haagerup H1 x Vec(C2)
2	R8_2_611	$\mathrm{TY}(C_2^2)^{S_3}$ with strict action
2	R8_2_839	Haagerup H1 x Fib

Five further multiplicity-one models are imported from exact AnyonWiki tensor matches: $R8_1_36$ is the crossed product of $\mathrm{TY}(C_3)$ by the order-two inversion action; $R8_1_61$ is the rank-eight fusion subcategory called one-half $SO(12)_2$; $R8_1_78$ is $SO(9)_2$; $R8_1_79$ is $Z(\mathrm{Vec}_{S_3})\simeq\mathrm{Rep}(D(S_3))$;

and $R8_1_85$ is the neutral component of the universal $\mathbb{Z}_2$ grading after gauging the full S_3 symmetry of $SO(8)_1$ [1, 2]. For the crossed product, the standard bicharacter $\chi(a,b) = \exp(2\pi iab/3)$ and positive $\tau = 1/\sqrt{3}$ are invariant under simultaneous inversion. The induced tensor action squares to the identity, and the crossed product is unitary. Its independently constructed tensor is matched in all 512 positions. The other four are standard unitary categories or fusion subcategories thereof; their complete sourced tensors and unit-preserving permutations are included in the positive catalogue. Existence of these models proves general complex categorifiability. No negative conclusion is imported merely from an AnyonWiki flag: the previously reported pentagon no-go for $R8_1_26$ is now replaced by the independent theoretical proof in Proposition 3.3.

4.1. **Twenty-two further constructions from extensions and subfactors.** The following constructions are verified by generating their multiplication tensors directly and comparing all 512 coefficients. No associator search is needed.

For $G = C_4$ and $G = C_2^2$, write $X_g = g\rho$ and use

$$gh = g + h, \qquad gX_h = X_{g+h}, \qquad X_g h = X_{g-h}, \qquad X_g X_h = (g-h) + m\sum_{a\in G} X_a.$$

At $m = 1$, Izumi's unitary generalized Haagerup constructions [8, Section 9] give $R8_1_15$ and $R8_1_69$, respectively. At $m = 2$, these are the Asaeda–Haagerup rings AH_4 and AH_5, giving $R8_2_39$ and $R8_2_484$. The C_4 construction is [9, Theorem 4.9]; existence of the C_2^2 category follows from its Lemma 3.15, Theorem 3.17 and Section 4. These are unitary subfactor categories.

The ring $R8_1_19$ is $(\mathrm{Fib}\boxtimes\mathrm{Fib})\rtimes C_2$, using the canonical action exchanging the two factors. The ring $R8_1_22$ is the fusion subcategory of $\mathrm{TY}(C_3)\boxtimes\mathrm{Vec}_{C_4}$ whose simple objects have matching parity: (a, z^{2j}), (m, z) and (m, z^3). Both operations preserve unitarity.

For $R8_1_30$, take the standard unitary $\mathrm{TY}(C_6)$ with bicharacter $\chi(a,b) = \exp(2\pi iab/6)$ and positive parameter $\tau = 1/\sqrt{6}$. Inversion fixes m, preserves every standard associator coefficient, and gives a tensor functor T with identity tensorator. Equip the C_2 action with coherence $T^2 \Rightarrow \mathrm{Id}$ equal to $+1$ on invertibles and -1 on m. This is a T-invariant monoidal natural automorphism, since the TY grading is $\mathbb{Z}/2$. In the equivariantization, the two structures on m have eigenvalues $\pm i$. The tensor decomposition is obtained by splitting the permutation action on each fusion space into its two characters; in particular two odd eigenvalues contribute the additional minus sign. The result is the stated full tensor, with four invertibles, two objects of dimension two, and two of dimension $\sqrt{6}$.

Let $F_{20} = C_5 \rtimes C_4$ with

$$(a,b)(c,d) = (a + 2^b c,\ b + d), \qquad \omega(x,y,z) = (-1)^{b(x)\lfloor (b(y)+b(z))/4\rfloor},$$

where the two coordinates are reduced modulo five and four, respectively, and representatives $0 \le b < 4$ are used in the cocycle. For $H = \langle (0,2)\rangle$, the restriction $\omega|_H$ is trivial. The unitary group-theoretical category $\mathcal{C}(F_{20}, \omega, H, 1)$ realizes $R8_1_3$. The supplied code constructs the eight simple homogeneous $\mathbb{C}[H]$-bimodules and their balanced tensor products. All phases are fourth roots of unity and are stored as integer exponents modulo four; the cocycle identity, bimodule coherence, balanced quotient, decomposition multiplicities, and full census tensor match are checked exactly.

The ring $R8_1_6$ has a bosonic quotient construction. Let $\mathcal{D} = PSU(2)_6$, with simples $1, f, \rho, f\rho$, where f is a fermion and $\rho^2 = 1 + \rho + f\rho$. Let $\mathcal{P} = \mathcal{C}(C_4, q)$ be the unitary pointed modular category with $q(z^a) = \exp(\pi i a^2/4)$. Then $b = f \boxtimes z^2$ in $\mathcal{B} = \mathcal{D}\boxtimes\mathcal{P}$ is an order-two boson: its self-braiding is $+1$. It generates a Tannakian $\mathrm{Rep}(C_2)$, whose regular algebra is $A = 1 \oplus b$. The central inclusion supplied by the braiding gives a fusion category $\mathcal{B}_A$ of all right A-modules, with tensor free-module functor F [21, Theorems 8.4.9 and 8.23.3]. The unitary crossed-product construction preserves its positive $*$-structure [39]. The boson is not transparent in $\mathcal{B}$; thus this uses all modules, rather than only local modules, and asserts no descended braiding.

Tensoring by b acts freely on the sixteen simples by $(V_j, z^a) \mapsto (V_{6-j}, z^{a+2})$, with $j \in \{0,2,4,6\}$. By adjunction every free module $F(V_j \boxtimes z^a)$ is simple, and two are isomorphic exactly when their

labels belong to the same orbit. Dominance of F gives precisely eight simples. With $g = F(1 \boxtimes z)$ and $r = F(\rho \boxtimes 1)$, their rules are

$$g^4 = 1, \qquad gr = rg, \qquad r^2 = 1 + r + g^2 r.$$

The ordered model basis $(1, g, g^2, g^3, r, gr, g^2r, g^3r)$ maps to census labels $(0, 1, 7, 2, 5, 3, 6, 4)$. The verifier reconstructs all coefficients from the truncated $SU(2)_6$ rules and the eight free orbits.

For *R*8_1_7, use either of the two unitary pseudo-$PSU(2)_6$ categories $\mathcal{D}$ of [29, Theorem 4.1]. Their basis $1, g, x, y$ obeys $g^2 = 1$, $gx = y$, $x^2 = y^2 = g + x + y$ and $xy = 1 + x + y$. The two source-bound AnyonWiki associator tables, labelled $(4, 1, 2, 4, 1, 0, 1)$ and $(4, 1, 2, 4, 2, 0, 1)$, are included in the supplement. Over $\mathbb{Q}(\zeta)$, $\zeta^4 + 1 = 0$, the verifier checks all 224 coefficients, 136 invertible associator blocks, unit and rigidity conditions and 2688 pentagons for each table.

Write $a(i) = gi$ and let $F_\ell^{ijk}[e, f]$ take left intermediate e to right intermediate f. In the two tables respectively, the scalars

$$(\lambda_1, \lambda_g, \lambda_x, \lambda_y) = (1, \zeta^2, \zeta, \zeta^3), \qquad (1, -\zeta^2, -\zeta^3, -\zeta)$$

define a half-braiding of g. The exact check is the following identity for each of the 24 fusion channels $N_{ij}^k = 1$:

$$\lambda_i \lambda_j F^{igj}_{a(k)}[a(i), a(j)] = \lambda_k F^{gij}_{a(k)}[a(i), k] F^{ijg}_{a(k)}[k, a(j)].$$

Thus $\widetilde{g} = (g, c)$ is an invertible object of $Z(\mathcal{D})$. Its self-braiding satisfies $\lambda_g^2 = -1$, so a fermionic condensation argument would be inappropriate. The equivalence between the unitary and ordinary centers permits a unitary representative of this lift [31, corrected Remark 3.1].

Instead take the trivial homomorphism $C_2 \to \mathrm{BrPic}(\mathcal{D})$ and modify its product equivalences by the normalized central 2-cocycle $t(1, 1) = \widetilde{g}$, with other normalized values the unit. The only nontrivial cocycle identity compares $\widetilde{g}$ with itself, regardless of its order in $\mathrm{Inv}\, Z(\mathcal{D})$. The remaining extension obstruction belongs to $H^4(C_2, \mathbb{C}^\times) = 0$ [7, Theorem 1.3]; this is also the trivial-extension zesting of [30, Section 6.2].

This extension is unitary. Choose regular unitary $\mathcal{D}$-bimodule components and unitary multiplication equivalences from the unitary lift. Preliminary associativity intertwiners can be chosen unitary by polar decomposition, so their normalized pentagon defect is $U(1)$-valued. A normalized 3-cochain on C_2 is determined by u at $(1, 1, 1)$, and its coboundary at $(1, 1, 1, 1)$ is u^2. A unit-modulus square root therefore removes the defect while keeping every coherence map unitary. The resulting regular-component extension is a unitary fusion category; this argument uses no general assertion that arbitrary extensions of unitary categories are unitarizable.

Its simples are (X, a), $a \in C_2$, with

$$(X, a)(Y, b) = (XY g^{ab}, a + b).$$

For $z = (1, 1)$, this gives $z^2 = g$, $zx = xz$ and $x^2 = z^2 + x + z^2 x$. The model basis $(1, g, x, y, z, zg, zx, zy)$ maps to census labels $(0, 7, 3, 4, 1, 2, 6, 5)$, and all 512 coefficients agree. The exact associator and central-lift replay is invoked by the finite-update verifier.

The same two rank-four bases have nontrivial flip extensions. For $\mathcal{D} = PSU(2)_6 = \mathrm{Ad}(A_7)$, a tensor autoequivalence τ exchanges ρ and $f\rho$ [32, Lemma 4.10]. Its square is tensor-isomorphic to the identity. The fusion rules imply $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{D}) = 1$, so this square isomorphism is unique after normalization and obeys the C_2 action coherence. Reutter's unitarization and automatic-unitarity theorems give a unitary representative of the entire action [33, Theorems 1–2].

For the second pseudo-$PSU(2)_6$ source table, the permutation $\tau = (x\ y)$ also gives a coherent tensor action, now checked explicitly. Its tensorator J_{ij}^k is 1 when an input is the unit. On the seventeen other admissible channels, ordered lexicographically, write $J = (\sqrt{2})^a \zeta^b$; the exponent

vectors are

$$a = (-1,-1,0,-1,-1,-1,-1,-1,0,-1,0,-1,0,-1,0,0,0),$$
$$b = (29,15,10,17,1,5,5,11,0,1,16,15,4,5,0,0,0).$$

All 224 tensor-functor identities hold over $\mathbb{Q}(\zeta)$. The natural isomorphism $\eta : \tau^2 \Rightarrow \mathrm{Id}$ has scalars

$$\eta = \left(1, \frac{1+\zeta^2}{2}, -\frac{1+\zeta^2}{2}, -\frac{1+\zeta^2}{2}\right)$$

in the order $(1, g, x, y)$. The verifier checks all 24 equations $\eta_i\eta_j = \eta_k J^k_{ij} J^{\tau(k)}_{\tau(i),\tau(j)}$ and $\eta_{\tau(i)} = \eta_i$. This is therefore a coherent action, not just a symmetry of the ring. Transport to unitary functors preserves its coherence; the transported monoidal natural isomorphism is automatically unitary by [33, Theorems 1–2].

For either rank-four base, its nontrivial invertible has a unique central lift: the kernel of $\mathrm{Inv}\, Z(\mathcal{D}) \to \mathrm{Inv}\, \mathcal{D}$ is $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{D}) = 1$, and a lift was already constructed above. The flip fixes this lift. Thus one may use either the crossed product or its central 2-cocycle modification. The remaining C_2 obstruction is removed by the unitary phase correction described above. With v the nontrivial invertible, their products are

$$(X,a)(Y,b) = \big(X\tau^a(Y)v^{\epsilon ab}, a+b\big), \qquad \epsilon \in \{0,1\}.$$

For $PSU(2)_6$, these give $R8_1_25$ at $\epsilon = 0$ and $R8_1_14$ at $\epsilon = 1$; for pseudo-$PSU(2)_6$, they give $R8_1_42$ and $R8_1_11$, respectively. The exact tensorator and coherent-square replay is `software/verify_pseudo_psu_swap.py`.

The ring $R8_1_23$ is $\mathrm{Vec}_{Q_8}^{C_3}$ for the strict automorphism $i \mapsto j \mapsto k \mapsto i$ of the quaternion group. The fixed objects $1, -1$ each have three equivariant structures, giving six invertibles. The two free orbits $\{i,j,k\}$ and $\{-i,-j,-k\}$ give two simples of dimension three. For a free output orbit, the fusion coefficient is the number of pairs of input group elements with product equal to its chosen representative. For a fixed output, the product fibre is a sum of free C_3-orbits, each contributing its regular character, together with any fixed pairs carrying the product character. These rules independently generate every coefficient. In the order of the three characters over 1, the three over -1, and the two free orbits, the census permutation is $(0,5,6,7,1,2,3,4)$.

The inversion construction for $R8_1_30$ also applies to $A = C_9$ and $A = C_3^2$, producing $R8_1_34$ and $R8_1_35$. Use respectively $\chi(a,b) = \exp(2\pi iab/9)$ and $\chi(a,b) = \exp(2\pi i(a_1b_1 + a_2b_2)/3)$, with $\tau = 1/3$. Inversion preserves these nondegenerate bicharacters, hence the standard unitary TY associators. The parity coherence is again -1 on m and $+1$ on group elements. There are two equivariant simples above 0, four free pairs of nonzero group elements, and two structures with eigenvalues $\pm i$ above m. Their dimensions are $1,1,2,2,2,2,3,3$; the same exact character projection generates the recorded tensors.

Finally take $A = \mathbb{F}_2^2$, $\chi(a,b) = (-1)^{a_1b_2+a_2b_1}$ and $\tau = 1/2$. The whole group $S_3 = \mathrm{GL}_2(\mathbb{F}_2)$ preserves χ, so label permutations give a strict unitary action on the standard TY presentation. For $t = 0, 1$, modify its action coherence on m by

$$\alpha_t(g,h) = (-1)^{tp(g)p(h)},$$

where $p : S_3 \to \mathbb{F}_2$ is parity, leaving the group simples unchanged. This is monoidal by the TY grading, and the action pentagon is the elementary normalized 2-cocycle identity for $p(g)p(h)$. Both equivariantizations are therefore unitary. Their orbits $\{0\}$, $A\backslash\{0\}$ and $\{m\}$ have stabilizers S_3, C_2, S_3. Denote their simples by A_ρ, X_ϵ and M_ρ, where ρ ranges over $1, \mathrm{sign}, \mathrm{std}$ and $\epsilon = \pm$. Above m, the equivariance is $i^{tp(g)}\rho(g)$, so the product of two such factors contributes sign^t.

The fusion coefficients are ordinary character inner products. Multiplication by A_ρ is tensoring with ρ, restricted to C_2 on an X-object. The other rules are

$$X_\epsilon X_\delta = A_{\mathrm{Ind}_{C_2}^{S_3}(\epsilon\delta)} + X_+ + X_-,$$
$$X_\epsilon M_\rho = M_{(\mathrm{Ind}_{C_2}^{S_3}\epsilon)\rho},$$
$$M_\rho M_\sigma = A_{\rho\sigma\,\mathrm{sign}^t} + X_{\mathrm{Res}_{C_2}^{S_3}(\rho\sigma\,\mathrm{sign}^t)},$$

where subscripts extend additively to reducible representations. For XX, the zero fibre is the induced character and the nonzero fibre is a regular C_2 permutation representation. These formulas give $R8_2_611$ at $t=0$ and $R8_2_154$ at $t=1$, both of global dimension 48. The model dimensions in the order $(A_1, A_{\mathrm{sign}}, A_{\mathrm{std}}, X_+, X_-, M_1, M_{\mathrm{sign}}, M_{\mathrm{std}})$ are $(1,1,2,3,3,2,2,4)$. All character arithmetic and tensor comparisons are exact.

One more quadratic model is $R8_1_53$. Yu–Zhang construct a unitary near-group category $NG(G,16)$ for $G=C_4^2$ [34, Proposition 3.1]. Its bicharacter and quadratic refinement may be taken as

$$\chi(g,h) = i^{g_1h_1-g_2h_2}, \qquad a(g) = \zeta_8^{3(g_1^2-g_2^2)}.$$

For $H=2G$, exact finite-group arithmetic gives $H=H^\perp$ and $a(h)=\chi(h,g_a)$ with $g_a=(1,1)\notin H$. Izumi's general orbifold theorem applies to $H\subseteq H^\perp$ even when $a|_H$ is nontrivial [35, Theorem 12.5]; its crossed-product-factor construction remains unitary. With $A=G/H=C_2^2$, the resulting rules are

$$\alpha_g\alpha_h=\alpha_{g+h}, \qquad \alpha_g\sigma=\sigma\alpha_g, \qquad \sigma^2=\alpha_{g_a}+\sum_{g\in A}\alpha_g\sigma.$$

There are four invertibles and four noninvertibles of dimension $(8+4\sqrt{5})/4=2+\sqrt{5}$, and every noninvertible has a distinct dual. This uses the non-self-dual theorem, rather than its self-dual Lagrangian corollary. In the model order $(00,01,10,11,\sigma,01\sigma,10\sigma,11\sigma)$, the census permutation is $(0,5,6,7,1,3,4,2)$. The portable verifier checks the subgroup hypotheses, the duality shift and all fusion coefficients; the cited theorem supplies categorical existence.

A cyclic version gives $R8_1_12$. Budinski constructs $NG(C_{16},16)$ [36, Theorem 3.1.1 and Table 3.1], with data

$$\chi(x,y)=\exp(2\pi i\,3xy/16), \qquad a(x)=\exp(-\pi i\,3x^2/16).$$

Although the table prints decimal arguments, the thesis's Section 3.3.2 gives an exact algebraic certification, with the defining-polynomial scope explicitly including $n=16$. We use that exact existence statement as a citation dependency; no numerical reconstruction is counted here as a category. For $H=4C_{16}$, one has $H=H^\perp$ and $a(h)=\chi(h,2)$. The same unitary orbifold theorem [35, Theorem 12.5] therefore gives

$$g^4=1, \qquad \sigma g=g^{-1}\sigma, \qquad \sigma^2=g^2+(1+g+g^2+g^3)\sigma.$$

In the order $(1,g,g^2,g^3,\sigma,g\sigma,g^2\sigma,g^3\sigma)$, the census permutation is $(0,1,7,2,3,6,4,5)$. The four noninvertibles are paired by duality, distinguishing this ring from the standard C_4 Haagerup–Izumi ring. The finite verifier checks all subgroup hypotheses and the full tensor exactly.

The ring $R8_1_83$ has a triality-gauging model. The S_3 gauging of the three-fermion category exists by [2, Section 6.2]; the unobstructed gauging data form the usual $H^3(S_3,U(1))$ torsor, including the twist denoted $m=0$ in [37, Section VI.D and Table IX]. This supplies an actual category. Starting from a pointed category, gauging produces a weakly group-theoretical fusion category, hence one admitting a unitary realization [38, Theorem 5.20]. Its eight integral simples, with dimensions $(1,1,2,3,3,4,4,4)$, form a fusion subcategory of dimension 72.

The integer fusion formulas for this subcategory are independently checked against the full twelve-object S-matrix of [37, Eq. (6.53), $m=0$]. Its entries lie in $\mathbb{Q}(\sqrt{2})$; using the unnormalized matrix, the verifier checks $S^2=144I$, all 1728 nonnegative integral Verlinde coefficients, and fusion closure

of the eight integral simples. It then compares the entire tensor with the pinned census under the stored permutation. Category existence comes from the gauging construction, not merely from the feasible modular datum.

The table provides one construction per positive tensor. Some have two isomorphic presentations; `data/realized_models.json` records every matched constructor, exact model tensor, precise model source and source-to-model isomorphism. Labels such as "pseudo-$PSU(2)_6$" denote the conventional fusion-ring name, not a weaker existence assertion: the cited rank-four table supplies a unitary category [14]. No undecided ring is in this table.

5. Reproduction and mathematical limits

The command `python3 software/analyze.py` reads the frozen source, checks its checksum and count, applies the ordered tests, and produces `data/decisions.json`, `data/exclusion_stages.csv`, `data/classification_counts.json` and `data/realized_models.json`. The command `python3 software/verify.py` checks source tensor identities, full based-ring uniqueness, all 973 statuses, every negative witness and every positive tensor match. It checks finite-module certificates by exhaustive integer reconstruction. The only native component is the rank-eight zero-spectrum witness search, compiled from `software/zero_spectrum.cpp`; the Python verifier rereads each nine-index witness and checks its conditions independently.

Both commands treat discovery failure, timeout, or a failed consistency check as an error, never as an exclusion. The separate signed campaign records completed certificates, inconclusive attempts and timeouts distinctly. The verifier independently regenerates every admitted rank-eight signed tensor, the inherited rank-seven negative proof trees, and all six appendix-based tensor transfers. Character-field calculations and subcategory dimension quotients use rational/algebraic arithmetic; no floating-point approximation gives a mathematical decision.

The finite update is independently replayed by `python3 software/verify_oct2.py`. It regenerates the twenty-two new model tensors, verifies the five grading obstructions and three solvability exclusions, and compares hashes of all 943 unaffected ledger records and all 38 previous positive records. The prior report is retained as `verification/pre_oct2_report.json`; the new report distinguishes this finite replay from the earlier full-certificate run.

This verification establishes the finite applications *conditional on* the cited theorems and the source census completeness. It does not settle the remaining 81 generally undecided rings, certify a multiplicity-three census, or classify categories up to tensor equivalence. Some criteria first exclude rings at multiplicity two, but nothing here implies that no new criterion is needed beyond that bound. Noncommutative Extended Haagerup models at rank eight have multiplicities four and seven [42]; they are genuine models but outside the present complete census bound.

AI disclosure. GPT-6 Astra assisted with literature exploration, mathematical research, code development, and manuscript preparation.

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